♠ .1 מצא פתרון של המשוואה $z^6 + 2z^2\bar{z} = 0$ המקיים את התנאים
$Re(z) > 0,\ \ Im(z) < 0.$

Since $z \neq 0$, the equation can be reduced to $z^4 + 2\bar{z} = 0$. Write it in the form $z^2 = -2\bar{z}$. This equation implies the following equations for $|z|$ and $Arg(z)$: $|z|^4 = 2|z|,\ \ 4Arg(z) = \pi - Arg(z)$. It follows $|z| = 2^{1/3}$,

$$5Arg(z) = \pi \iff Arg(z) = \frac{\pi + 2\pi k}{5}, \quad k = 0, 1, 2, 3, 4.$$

The requirement $Re(z) > 0,\ Im(z) < 0$ holds only for $k = 4$. The argument for $k = 4$ is the same as for $k = -1$. Therefore within given requirement the solution is unique: $z = 2^{1/3}(cos(\pi/5) - i \cdot sin(\pi/5))$.

♠ .2 מצא דוגמא של פולינום $P(z)$, לא פולינום אפס,
עם מקדמים ממשיים כך שלמערכת $P(z) = P'(z) = 0$ יש פתרון $z = z_0$.

In each of the versions z_0 is a given complex number with non-zero imaginary part.

The given equations imply that z_0 is a root of multiplicity (RIBUY) ≥ 2. The fact that $P(z)$ has real coefficients implies that $\bar{z}_0$ is also a root of multiplicity ≥ 2. Therefore the simplest example is

$$P(z) = (z - z_0)^2(z - \bar{z}_0)^2 = ((z - z_0)(z - \bar{z}_0))^2 = (z^2 - 2Re(z_0) + |z_0|^2)^2$$

and it remains to substitute to here the given z_0. For example for $z_0 = 3 - 2i$ we obtain $P(z) = (z^2 - 6z + 13)^2$.

♠ .3 מצא $a, b \in \mathbf{C}$ כך שלמערכת המשוואות הליניאריות
$\begin{pmatrix} a & 1 & 0 \\ 0 & 1 & 2 \\ 1+i & 0 & 1 \end{pmatrix} \cdot z = \begin{pmatrix} 1 \\ b \\ 0 \end{pmatrix}$ יש אינסוף פתרונות $z \in \mathbf{C}^3$.

In some versions a slightly different matrix and/or right hand side was given. In any version the solution is a regular dirug of a matrix, simple work with complex numbers.

♠ **4.** תהי A מטריצה ממשית 3×4 ו־ $\tilde{b} \in \mathbf{R}^3$.

נתון:

1. למערכת המשוואות הליניאריות $Ax = \tilde{b}$ יש פתרונות

$$\begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} a \\ b \\ c \\ 2 \end{pmatrix}.$$

2. $rankA = 3$.

מצא $a, b \in \mathbf{R}$.

In some versions the vector $\begin{pmatrix} a \\ b \\ c \\ 2 \end{pmatrix}$ was replaced by $\begin{pmatrix} 2 \\ a \\ b \\ c \end{pmatrix}$ or another vector with one of the coordinates a number and the other coordinates – parameters. The way of solution is exactly the same.

Denote the given vectors by v_1, v_2, v_3. The vectors $v_2 - v_1$ and $v_3 - v_1$ are solutions of the system $Ax = 0$. None of these vectors is zero vector. Since $rankA = 3$ any two non-zero solutions of the system $Ax = 0$ must be proportional. Therefore the vectors $v_2 - v_1 = \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \end{pmatrix}, \quad v_3 - v_1 = \begin{pmatrix} a-1 \\ b-2 \\ c-1 \\ 2 \end{pmatrix}$ are proportional, i.e. $\begin{pmatrix} a-1 \\ b-2 \\ c-1 \\ 2 \end{pmatrix} = k \cdot \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$ for some k. The equation for the last coordinate implies $k = 2$ and it follows $a = -1, b = 0, c = -1$.

♠ .5 יהיו

$$A = \text{ given matrix with a parameter } a, \quad B = A^4 - A^3.$$

לאיזה $a \in \mathbf{R}$ המטריצה B היא הפיכה?

---

the matrix B is invertible if and only if $detB \neq 0$. One has: $detB = det(A^4 - A^3) = det(A^3(A-I)) = (detA)^3 \cdot det(A-I)$, therefore B is invertible if and only if $detA \neq 0$ and $det(A-I) \neq 0$. It remains to compute the determinantes – a simple job.

♠ .6 הוכח שאם A ו־ B המטריצות 5×5 כך ש־

$$AB + BA = 0$$

אז לפחות אחת משתי המטריצות A, B אינה הפיכה.

עזרה: אם C מטרחצה ריבועת אז $det(-C) = ?$

$AB + BA = 0 \Rightarrow AB = -BA \Rightarrow det(AB) = det(-BA) \Rightarrow det(AB) = (-1)^5 det(BA) \Rightarrow det(A) \cdot det(B) = -(detA) \cdot det(B) \Rightarrow det(A) \cdot det(B) = 0.$

Therefore at least one of the determinates $det(A), det(B)$ is equal to 0 and consequently at least one of the matrices A, B is not invertible.